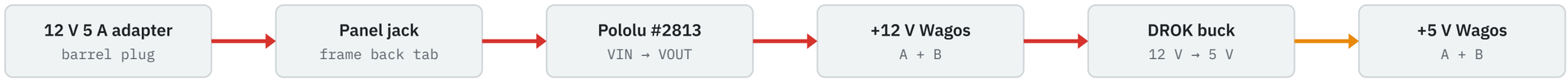
Ball balancer wiring

Every connection, generated from the same list as the tables below (`docs/wiring.py`). Power and ground run through eight WAGO 221-415 lever connectors: two for +12 V, two for +5 V, four for ground, each pair or group linked by a short jumper. Signals are Dupont leads. Hover a wire to see its two ends.

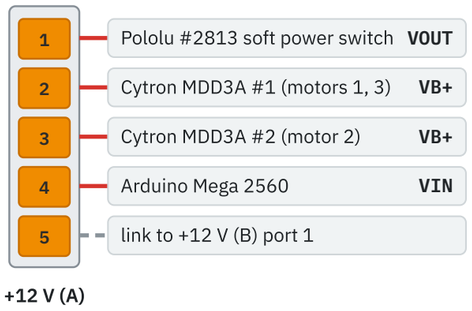
+12 V · +5 V · GND · Wago link · motor + · motor – · encoder A · encoder B · PWM · touch panel · buttons · power-off control

Power: what plugs into every Wago port

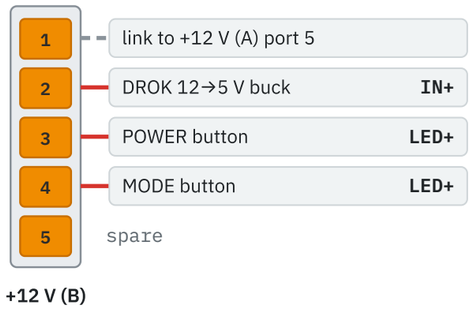


Every ground goes to the GND Wagos below. Linked Wagos act as one bus: the dashed port joins them.

+12 V bus

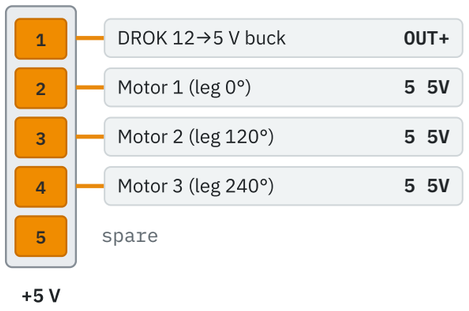


+12 V (A)



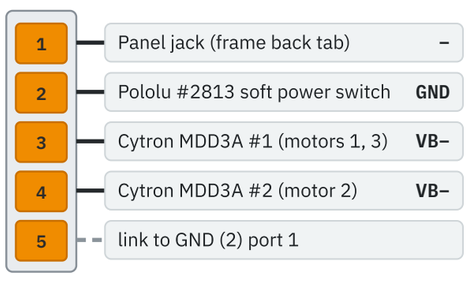
+12 V (B)

+5 V bus

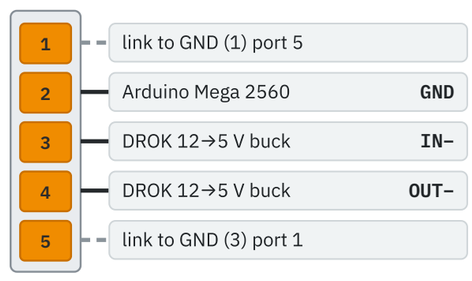


+5 V

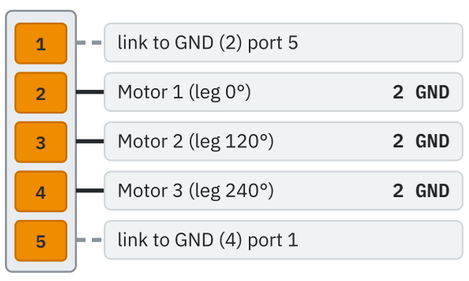
Ground bus



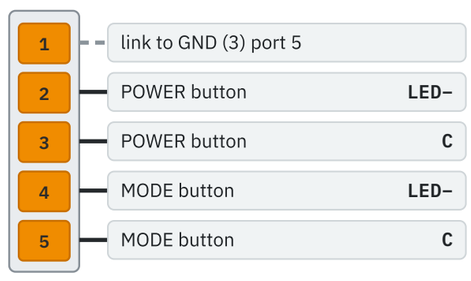
GND (1)



GND (2)

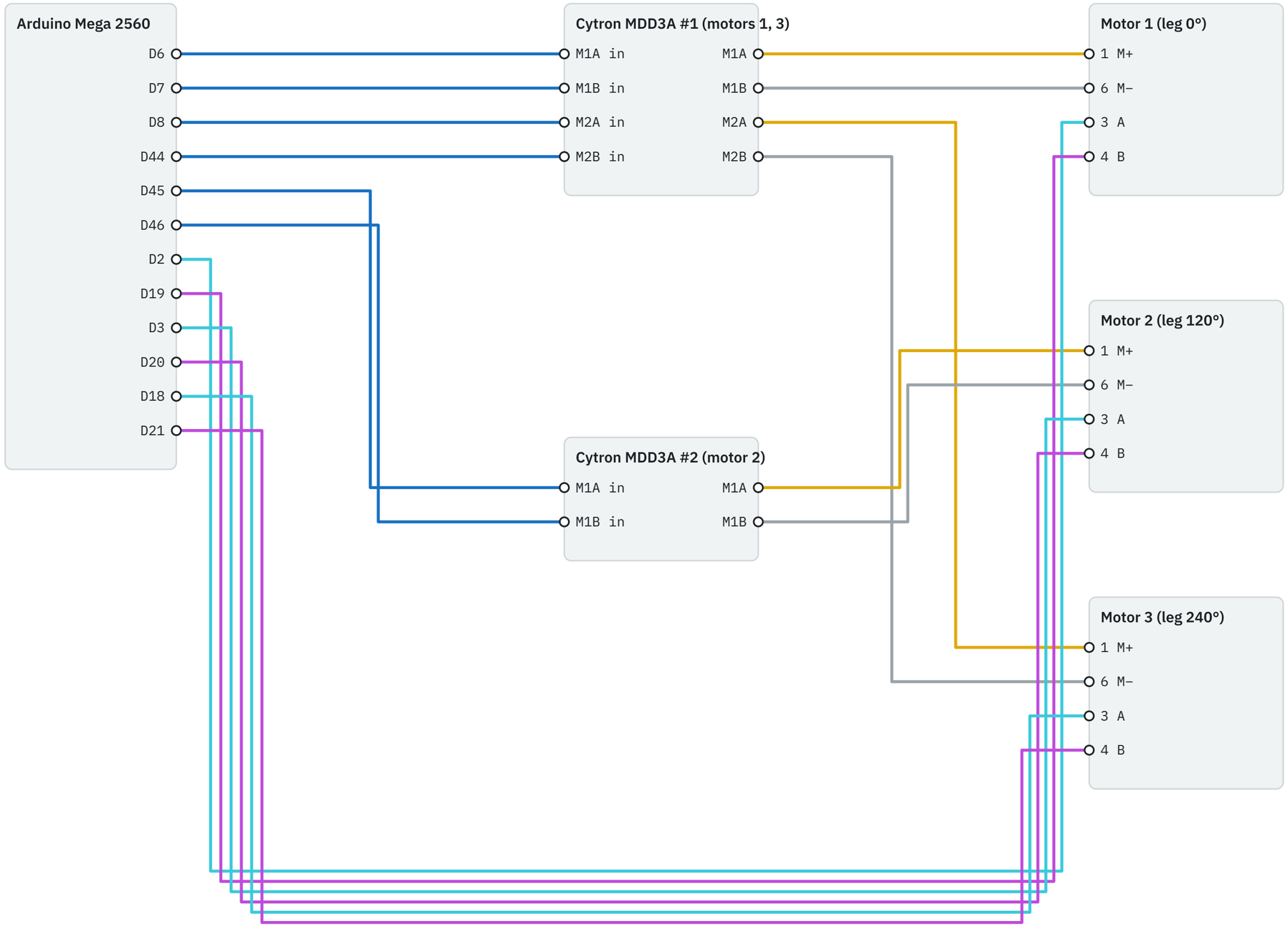


GND (3)

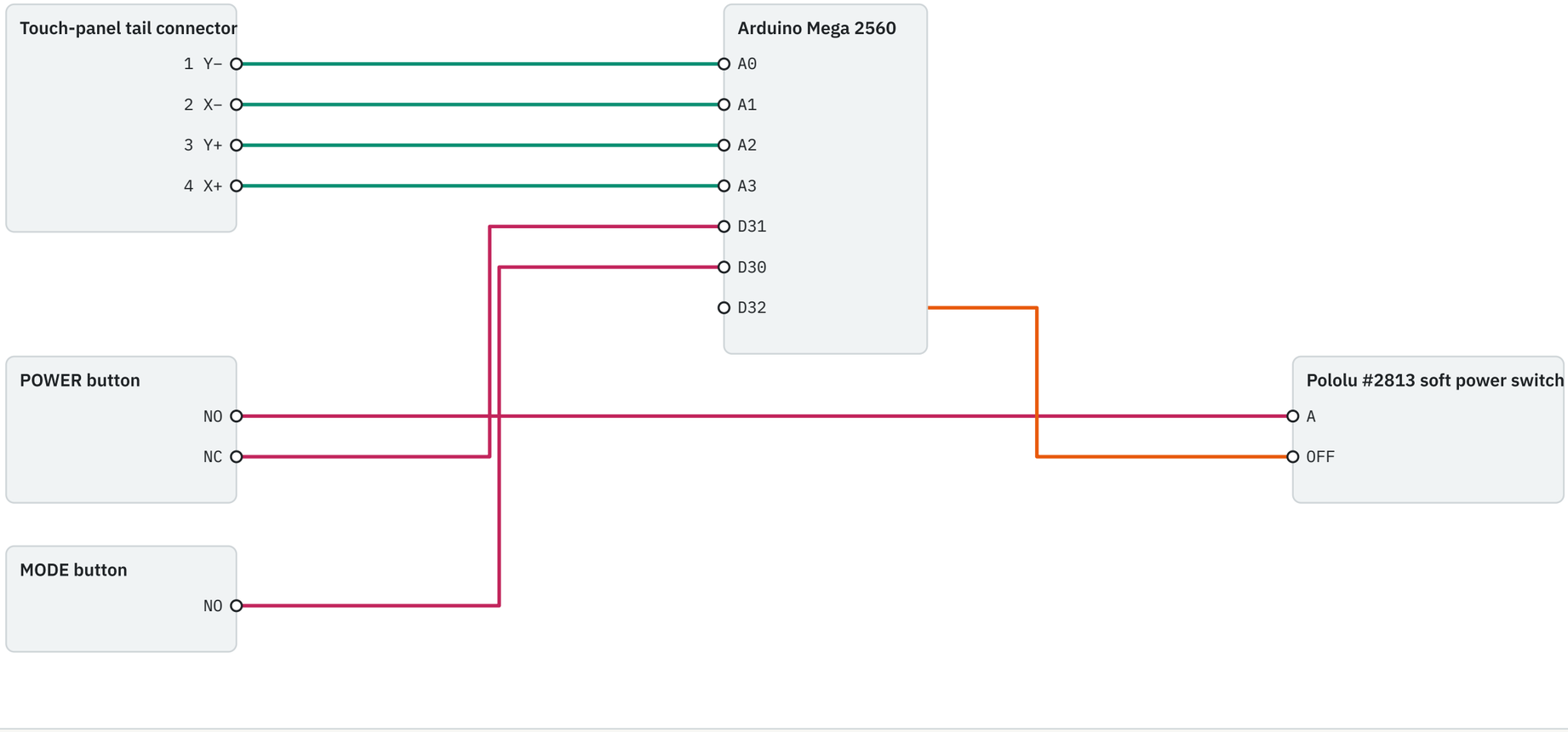


GND (4)

Motors: control signals, motor power and encoders



Touch panel, buttons and soft power



Before you power it on

- Fit the motors' leads last. First power-up: no motors connected, check 12 V on the +12 V Wagos and 5.0 V on the +5 V Wagos with a meter.
- Never connect the buck's 5 V output and the Mega's USB 5 V together; the Mega takes 12 V on VIN only and makes its own 5 V.
- Touch-panel tail order (Y−, X−, Y+, X+) is from a similar panel's datasheet: confirm with a meter (about 200–900 Ω across each X pair and each Y pair).
- Motor connector pins, left to right: 1 M+, 2 GND, 3 A, 4 B, 5 5V, 6 M−. Port labels give the pin number.
- Our motor cable colours: M+ yellow, GND black, A blue, B purple, 5V red, M− white.
- Our button harness colours: NO blue, C green, NC yellow, LED+ red, LED− black.
- Button harness colours vary by batch: identify LED+, LED−, C, NO and NC with a meter before connecting.
- Program the Mega over USB with the shell lifted; the soft switch can stay off while USB powers the Mega's logic.
- Second MDD3A's channel 2 is unused. Leave both MDD3As' 5V pins unconnected: they are outputs from the board's own regulator.

Wire list by part

Pololu #2813 soft power switch

PIN	GOES TO	WIRE
A	POWER button · NO	button harness to Dupont
GND	Wago GND (1) · 2	black, 20 AWG
OFF	Arduino Mega 2560 · D32	Dupont
VIN	Panel jack (frame back tab) · +	red, 20 AWG
VOUT	Wago +12 V (A) · 1	red, 20 AWG

Panel jack (frame back tab)

PIN	GOES TO	WIRE
+	Pololu #2813 soft power switch · VIN	red, 20 AWG
–	Wago GND (1) · 1	black, 20 AWG
socket	12 V 5 A wall adapter · plug	barrel plug

DROK 12→5 V buck

PIN	GOES TO	WIRE
IN+	Wago +12 V (B) · 2	red, 20 AWG
IN–	Wago GND (2) · 3	black, 20 AWG
OUT+	Wago +5 V · 1	red/orange, 20 AWG
OUT–	Wago GND (2) · 4	black, 20 AWG

Arduino Mega 2560

PIN	GOES TO	WIRE
A0	Touch-panel tail connector · 1 Y−	Dupont
A1	Touch-panel tail connector · 2 X−	Dupont
A2	Touch-panel tail connector · 3 Y+	Dupont
A3	Touch-panel tail connector · 4 X+	Dupont
D18	Motor 3 (leg 240°) · 3 A	motor pin 3, A (blue) to Dupont
D19	Motor 1 (leg 0°) · 4 B	motor pin 4, B (purple) to Dupont
D2	Motor 1 (leg 0°) · 3 A	motor pin 3, A (blue) to Dupont
D20	Motor 2 (leg 120°) · 4 B	motor pin 4, B (purple) to Dupont
D21	Motor 3 (leg 240°) · 4 B	motor pin 4, B (purple) to Dupont
D3	Motor 2 (leg 120°) · 3 A	motor pin 3, A (blue) to Dupont
D30	MODE button · NO	button harness to Dupont
D31	POWER button · NC	button harness to Dupont
D32	Pololu #2813 soft power switch · OFF	Dupont
D44	Cytron MDD3A #1 (motors 1, 3) · M2B in	Dupont
D45	Cytron MDD3A #2 (motor 2) · M1A in	Dupont
D46	Cytron MDD3A #2 (motor 2) · M1B in	Dupont
D6	Cytron MDD3A #1 (motors 1, 3) · M1A in	Dupont
D7	Cytron MDD3A #1 (motors 1, 3) · M1B in	Dupont
D8	Cytron MDD3A #1 (motors 1, 3) · M2A in	Dupont

Cytron MDD3A #1 (motors 1, 3)

PIN	GOES TO	WIRE
M1A	Motor 1 (leg 0°) · 1 M+	motor pin 1, M+ (yellow)
M1B	Motor 1 (leg 0°) · 6 M−	motor pin 6, M− (white)
M2A	Motor 3 (leg 240°) · 1 M+	motor pin 1, M+ (yellow)
M2B	Motor 3 (leg 240°) · 6 M−	motor pin 6, M− (white)
VB+	Wago +12 V (A) · 2	red, 20 AWG
VB−	Wago GND (1) · 3	black, 20 AWG
M1A in	Arduino Mega 2560 · D6	Dupont
M1B in	Arduino Mega 2560 · D7	Dupont
M2A in	Arduino Mega 2560 · D8	Dupont
M2B in	Arduino Mega 2560 · D44	Dupont

Cytron MDD3A #2 (motor 2)

PIN	GOES TO	WIRE
M1A	Motor 2 (leg 120°) · 1 M+	motor pin 1, M+ (yellow)
M1B	Motor 2 (leg 120°) · 6 M−	motor pin 6, M− (white)
VB+	Wago +12 V (A) · 3	red, 20 AWG
VB−	Wago GND (1) · 4	black, 20 AWG
M1A in	Arduino Mega 2560 · D45	Dupont
M1B in	Arduino Mega 2560 · D46	Dupont

PIN	GOES TO	WIRE
GND	Wago GND (2) · 2	black, 20 AWG
VIN	Wago +12 V (A) · 4	red, 20 AWG

Motor 1 (leg 0°)

PIN	GOES TO	WIRE
3 A	Arduino Mega 2560 · D2	motor pin 3, A (blue) to Dupont
4 B	Arduino Mega 2560 · D19	motor pin 4, B (purple) to Dupont
1 M+	Cytron MDD3A #1 (motors 1, 3) · M1A	motor pin 1, M+ (yellow)
2 GND	Wago GND (3) · 2	black, 20 AWG
5 5V	Wago +5 V · 2	red/orange, 20 AWG
6 M–	Cytron MDD3A #1 (motors 1, 3) · M1B	motor pin 6, M– (white)

Motor 2 (leg 120°)

PIN	GOES TO	WIRE
3 A	Arduino Mega 2560 · D3	motor pin 3, A (blue) to Dupont
4 B	Arduino Mega 2560 · D20	motor pin 4, B (purple) to Dupont
1 M+	Cytron MDD3A #2 (motor 2) · M1A	motor pin 1, M+ (yellow)
2 GND	Wago GND (3) · 3	black, 20 AWG
5 5V	Wago +5 V · 3	red/orange, 20 AWG
6 M–	Cytron MDD3A #2 (motor 2) · M1B	motor pin 6, M– (white)

Motor 3 (leg 240°)

PIN	GOES TO	WIRE
3 A	Arduino Mega 2560 · D18	motor pin 3, A (blue) to Dupont
4 B	Arduino Mega 2560 · D21	motor pin 4, B (purple) to Dupont
1 M+	Cytron MDD3A #1 (motors 1, 3) · M2A	motor pin 1, M+ (yellow)
2 GND	Wago GND (3) · 4	black, 20 AWG
5 5V	Wago +5 V · 4	red/orange, 20 AWG
6 M–	Cytron MDD3A #1 (motors 1, 3) · M2B	motor pin 6, M– (white)

Touch-panel tail connector

PIN	GOES TO	WIRE
1 Y–	Arduino Mega 2560 · A0	Dupont
2 X–	Arduino Mega 2560 · A1	Dupont
3 Y+	Arduino Mega 2560 · A2	Dupont
4 X+	Arduino Mega 2560 · A3	Dupont

POWER button

PIN	GOES TO	WIRE
C	Wago GND (4) · 3	black, 20 AWG
NC	Arduino Mega 2560 · D31	button harness to Dupont
NO	Pololu #2813 soft power switch · A	button harness to Dupont
LED+	Wago +12 V (B) · 3	red, 20 AWG
LED–	Wago GND (4) · 2	black, 20 AWG

MODE button

PIN	GOES TO	WIRE
C	Wago GND (4) · 5	black, 20 AWG
NO	Arduino Mega 2560 · D30	button harness to Dupont
LED+	Wago +12 V (B) · 4	red, 20 AWG
LED–	Wago GND (4) · 4	black, 20 AWG

Wago +12 V (A)

PIN	GOES TO	WIRE
1	Pololu #2813 soft power switch · VOUT	red, 20 AWG
2	Cytron MDD3A #1 (motors 1, 3) · VB+	red, 20 AWG
3	Cytron MDD3A #2 (motor 2) · VB+	red, 20 AWG
4	Arduino Mega 2560 · VIN	red, 20 AWG
5	Wago +12 V (B) · 1	short jumper, same colour as its bus

Wago +12 V (B)

PIN	GOES TO	WIRE
1	Wago +12 V (A) · 5	short jumper, same colour as its bus
2	DROK 12→5 V buck · IN+	red, 20 AWG
3	POWER button · LED+	red, 20 AWG
4	MODE button · LED+	red, 20 AWG

Wago +5 V

PIN	GOES TO	WIRE
1	DROK 12→5 V buck · OUT+	red/orange, 20 AWG
2	Motor 1 (leg 0°) · 5 5V	red/orange, 20 AWG
3	Motor 2 (leg 120°) · 5 5V	red/orange, 20 AWG
4	Motor 3 (leg 240°) · 5 5V	red/orange, 20 AWG

Wago GND (1)

PIN	GOES TO	WIRE
1	Panel jack (frame back tab) · –	black, 20 AWG
2	Pololu #2813 soft power switch · GND	black, 20 AWG
3	Cytron MDD3A #1 (motors 1, 3) · VB–	black, 20 AWG
4	Cytron MDD3A #2 (motor 2) · VB–	black, 20 AWG
5	Wago GND (2) · 1	short jumper, same colour as its bus

Wago GND (2)

PIN	GOES TO	WIRE
1	Wago GND (1) · 5	short jumper, same colour as its bus
2	Arduino Mega 2560 · GND	black, 20 AWG
3	DROK 12→5 V buck · IN–	black, 20 AWG
4	DROK 12→5 V buck · OUT–	black, 20 AWG
5	Wago GND (3) · 1	short jumper, same colour as its bus

Wago GND (3)

PIN	GOES TO	WIRE
1	Wago GND (2) · 5	short jumper, same colour as its bus
2	Motor 1 (leg 0°) · 2 GND	black, 20 AWG
3	Motor 2 (leg 120°) · 2 GND	black, 20 AWG
4	Motor 3 (leg 240°) · 2 GND	black, 20 AWG
5	Wago GND (4) · 1	short jumper, same colour as its bus

Wago GND (4)

PIN	GOES TO	WIRE
1	Wago GND (3) · 5	short jumper, same colour as its bus
2	POWER button · LED–	black, 20 AWG
3	POWER button · C	black, 20 AWG
4	MODE button · LED–	black, 20 AWG
5	MODE button · C	black, 20 AWG